



ULUSLARARASI KIBRIS ÜNİVERSİTESİ
CYPRUS INTERNATIONAL UNIVERSITY

Car Rental Management System

CMPE343 DBMS-I: Term Project

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1.Introduction:

The Car Rental Management System is designed to manage car inventory, customer bookings, rental transactions, and payment details. It allows the rental company to handle car availability, pricing based on categories, and customer reservations efficiently.

Assumptions:

- Each car has a unique plate number.
- Cars belong to specific categories (e.g., Sedan, SUV) which determine the daily rental rate.
- Customers can rent multiple cars over time.
- Payments are linked to specific rental agreements.
- Staff members manage the rental process.

2.Database Design:

2.1 ER Diagram

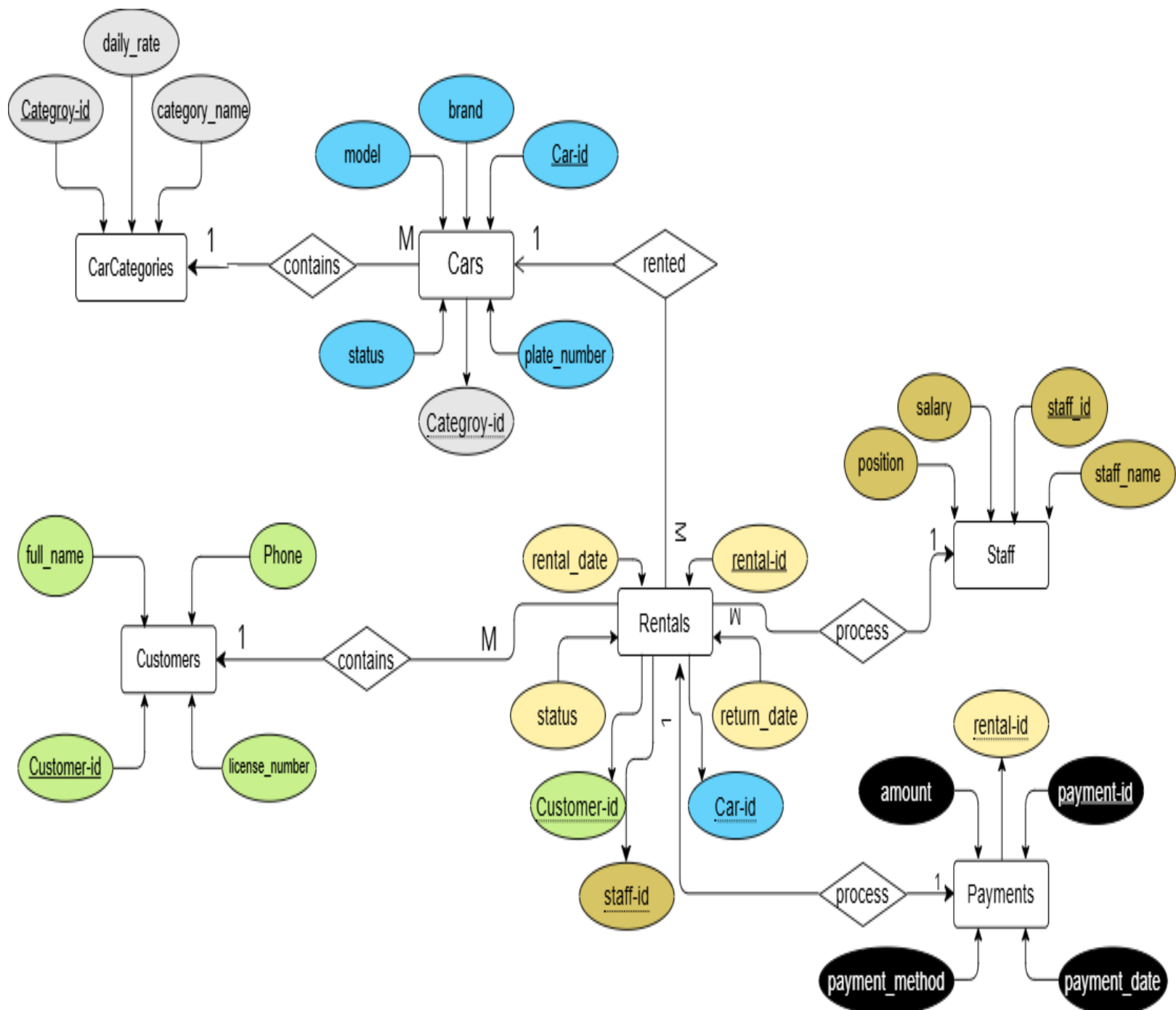
- **Entities:**

- CarCategories
- Cars
- Customers
- Staff
- Rentals
- Payments

- **Relationships:**

- One Category includes multiple Cars.
- One Customer can make multiple Rentals.
- One Staff member can process multiple Rentals.
- Each Rental is linked to one Car and one Payment

ER Diagram:



2.2 Creating Tables & Relations

1. CarCategories:

```
1
2 CREATE TABLE CarCategories (
3     category_id INT PRIMARY KEY,
4     category_name VARCHAR(50) NOT NULL,
5     daily_rate DECIMAL(10, 2) NOT NULL
6 );
7
```

2. Staff:

```
8 ✓ CREATE TABLE Staff (
9     staff_id INT PRIMARY KEY,
10    staff_name VARCHAR(100) NOT NULL,
11    position VARCHAR(50),
12    salary DECIMAL(10, 2)
13 );
```

3. Customers:

```
14 ✓ CREATE TABLE Customers (
15     customer_id INT PRIMARY KEY,
16     full_name VARCHAR(100) NOT NULL,
17     license_number VARCHAR(50) UNIQUE NOT NULL,
18     phone VARCHAR(20)
19 );
```

4. Cars

```
20
21 ✓ CREATE TABLE Cars (
22     car_id INT PRIMARY KEY,
23     brand VARCHAR(50) NOT NULL,
24     model VARCHAR(50) NOT NULL,
25     year INT,
26     plate_number VARCHAR(20) UNIQUE NOT NULL,
27     status VARCHAR(20) DEFAULT 'Available',
28     category_id INT,
29     FOREIGN KEY (category_id) REFERENCES CarCategories
    (category_id)
30 );
```

5. Rentals:

```
31
32 CREATE TABLE Rentals (
33     rental_id INT PRIMARY KEY,
34     car_id INT NOT NULL,
35     customer_id INT NOT NULL,
36     staff_id INT NOT NULL,
37     rental_date DATE NOT NULL,
38     return_date DATE,
39     status VARCHAR(20) DEFAULT 'Active',
40     FOREIGN KEY (car_id) REFERENCES Cars(car_id),
41     FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
42     FOREIGN KEY (staff_id) REFERENCES Staff(staff_id)
43 );
44
```

6. Payments:

```
45 ✓ CREATE TABLE Payments (  
46     payment_id INT PRIMARY KEY,  
47     rental_id INT NOT NULL,  
48     amount DECIMAL(10, 2) NOT NULL,  
49     payment_date DATE NOT NULL,  
50     payment_method VARCHAR(50),  
51     FOREIGN KEY (rental_id) REFERENCES Rentals(rental_id)  
52 );
```

3 Populating Data

1. CarCategories:

```
1  
2 INSERT INTO CarCategories (category_id, category_name,  
   daily_rate) VALUES  
3 (1, 'Sedan', 50.00),  
4 (2, 'SUV', 85.00),  
5 (3, 'Luxury', 250.00),  
6 (4, 'Hatchback', 40.00),  
7 (5, 'Sports', 300.00),  
8 (6, 'Van', 100.00);  
9
```

2. Staff:

```
10 INSERT INTO Staff (staff_id, staff_name, position, salary) VALUES
11 (1, 'Ali Hassan', 'Manager', 15000),
12 (2, 'Sara Ahmed', 'Receptionist', 5500),
13 (3, 'Omar Khaled', 'Sales Agent', 6200),
14 (4, 'Layla Mohammed', 'Accountant', 8000),
15 (5, 'Fahad Ibrahim', 'Sales Agent', 6000),
16 (6, 'Mona Ali', 'Cleaner', 3000);
17
```

3. Customers:

```
18 INSERT INTO Customers (customer_id, full_name, license_number,
19 phone) VALUES
20 (101, 'Sarah Nasser', 'LIC-99901', '0551234567'),
21 (102, 'Sufyan Rashad', 'LIC-88802', '0509876543'),
22 (103, 'Majd Nasr', 'LIC-77703', '0561112233'),
23 (104, 'Ramez Al-Omari', 'LIC-66604', '0544445566'),
24 (105, 'Omar Al-Kaff', 'LIC-55505', '0533332211'),
25 (106, 'Laila Al-Harbi', 'LIC-44406', '0599988776'),
26 (107, 'Zaid Al-Sultan', 'LIC-33307', '0511122233'),
27 (108, 'Hanan Mansour', 'LIC-22208', '0522233344'),
28 (109, 'Fahad Bin Khalid', 'LIC-11109', '0588877766'),
29 (110, 'Noura Al-Said', 'LIC-00010', '0577766655');
```

4. Cars:

```
30 INSERT INTO Cars (car_id, brand, model, year, plate_number,
31 status, category_id) VALUES
32 (501, 'Toyota', 'Camry', 2022, 'ABC-123', 'Rented', 1),
33 (502, 'Hyundai', 'Tucson', 2023, 'XYZ-987', 'Available', 2),
34 (503, 'BMW', 'Series 7', 2021, 'LUX-001', 'Maintenance', 3),
35 (504, 'Honda', 'Accord', 2020, 'HND-555', 'Available', 1),
36 (505, 'Mercedes', 'S-Class', 2024, 'MER-111', 'Rented', 3),
37 (506, 'Nissan', 'Patrol', 2022, 'NIS-000', 'Available', 2),
38 (507, 'Kia', 'Rio', 2021, 'KIA-444', 'Available', 4),
39 (508, 'Ferrari', '488', 2023, 'SPD-999', 'Available', 5),
40 (509, 'Ford', 'Transit', 2019, 'VAN-777', 'Available', 6),
41 (510, 'Toyota', 'Corolla', 2022, 'T-882', 'Available', 1);
```


5. Rentals:

```
42  INSERT INTO Rentals (rental_id, car_id, customer_id, staff_id,  
    rental_date, return_date, status) VALUES  
43  (1, 501, 101, 2, '2025-12-01', '2025-12-05', 'Completed'),  
44  (2, 502, 102, 3, '2025-12-10', NULL, 'Active'),  
45  (3, 505, 103, 5, '2025-12-12', '2025-12-15', 'Completed'),  
46  (4, 501, 104, 3, '2025-12-16', '2025-12-18', 'Completed'),  
47  (5, 506, 105, 2, '2025-12-20', NULL, 'Active'),  
48  (6, 504, 106, 3, '2025-12-21', '2025-12-25', 'Completed'),  
49  (7, 503, 101, 5, '2025-12-26', NULL, 'Active'),  
50  (8, 508, 107, 3, '2025-12-27', '2025-12-29', 'Completed');  
51
```

6. Payments:

```
52  INSERT INTO Payments (payment_id, rental_id, amount,  
    payment_date, payment_method) VALUES  
53  (1, 1, 250.00, '2025-12-05', 'Credit Card'),  
54  (2, 3, 750.00, '2025-12-15', 'Cash'),  
55  (3, 4, 100.00, '2025-12-18', 'Debit Card'),  
56  (4, 6, 200.00, '2025-12-25', 'Credit Card'),  
57  (5, 8, 600.00, '2025-12-29', 'Apple Pay');
```

4 Display Tables

1. CarCategories:

 catego... i... ▼	category_name varchar ▼	daily_rate numeric ▼
1	Sedan	50.00
2	SUV	85.00
3	Luxury	250.00
4	Hatchback	40.00
5	Sports	300.00
6	Van	100.00

2. Staff:

 sta... i... ▼	staff_name varchar ▼	position varchar ▼	sal... nume... ▼
1	Ali Hassan	Manager	15000.00
2	Sara Ahmed	Receptionist	5500.00
3	Omar Khaled	Sales Agent	6200.00
4	Layla Mohammed	Accountant	8000.00
5	Fahad Ibrahim	Sales Agent	6000.00
6	Mona Ali	Cleaner	3000.00

3. Customers:

 custo... i... ▾	full_name varchar ▾	license_number varchar ▾	phone varchar ▾
101	Sarah Nasser	LIC-99901	0551234567
102	Sufyan Rashad	LIC-88802	0509876543
103	Majd Nasr	LIC-77703	0561112233
104	Ramez Al-Omari	LIC-66604	0544445566
105	Omar Al-Kaff	LIC-55505	0533332211
106	Laila Al-Harbi	LIC-44406	0599988776
107	Zaid Al-Sultan	LIC-33307	0511122233
108	Hanan Mansour	LIC-22208	0522233344
109	Fahad Bin Khalid	LIC-11109	0588877766
110	Noura Al-Said	LIC-00010	0577766655

4. Cars:

 car... i... ▾	brand varchar ▾	model varchar ▾	year int4 ▾	plate_number varchar ▾	status varchar ▾	 catego... i... ▾
501	Toyota	Camry	2022	ABC-123	Rented	1 
502	Hyundai	Tucson	2023	XYZ-987	Available	2 
503	BMW	Series 7	2021	LUX-001	Maintenance	3 
504	Honda	Accord	2020	HND-555	Available	1 
505	Mercedes	S-Class	2024	MER-111	Rented	3 
506	Nissan	Patrol	2022	NIS-000	Available	2 
507	Kia	Rio	2021	KIA-444	Available	4 
508	Ferrari	488	2023	SPD-999	Available	5 
509	Ford	Transit	2019	VAN-777	Available	6 
510	Toyota	Corolla	2022	T-882	Available	1 

5. Rentals:

 rent... i... ▾	 car... i... ▾	 custo... i... ▾	 sta... i... ▾	rental_date date ▾	return_date date ▾	status varchar ▾
1	501 	101 	2 	2025-12-01	2025-12-05	Completed
2	502 	102 	3 	2025-12-10	NULL	Active
3	505 	103 	5 	2025-12-12	2025-12-15	Completed
4	501 	104 	3 	2025-12-16	2025-12-18	Completed
5	506 	105 	2 	2025-12-20	NULL	Active
6	504 	106 	3 	2025-12-21	2025-12-25	Completed
7	503 	101 	5 	2025-12-26	NULL	Active
8	508 	107 	3 	2025-12-27	2025-12-29	Completed

6. Payments:

paym... i... ▾	rent... i... ▾	amo... num... ▾	payment_d... d... ▾	payment_method varchar ▾
1	1 →	250.00	2025-12-05	Credit Card
2	3 →	750.00	2025-12-15	Cash
3	4 →	100.00	2025-12-18	Debit Card
4	6 →	200.00	2025-12-25	Credit Card
5	8 →	600.00	2025-12-29	Apple Pay

5 Queries:

1. Display customer names and format phone numbers with an international code.

```
1 SELECT full_name, ('+966-' || phone) AS International_Phone FROM Customers;
```

Output:

full_name	international_phone
Sarah Nasser	+966-0551234567
Sufyan Rashad	+966-0509876543
Majd Nasr	+966-0561112233
Ramez Al-Omari	+966-0544445566
Omar Al-Kaff	+966-0533332211
Laila Al-Harbi	+966-059988776
Zaid Al-Sultan	+966-0511122233
Hanan Mansour	+966-0522233344
Fahad Bin Khalid	+966-0588877766
Noura Al-Said	+966-0577766655

2. Display the car brand and model concatenated in uppercase.

```
1 SELECT UPPER(brand || ' ' || model) AS Car_Full_Name FROM Cars;
```

Output:

car_full_name
TOYOTA CAMRY
HYUNDAI TUCSON
BMW SERIES 7
HONDA ACCORD
MERCEDES S-CLASS
NISSAN PATROL
KIA RIO
FERRARI 488
FORD TRANSIT
TOYOTA COROLLA

3. Calculate the number of days remaining until the end of the year 2025 for each rental.

```
1 SELECT rental_id, rental_date, ('2025-12-31'::date - rental_date)
   AS Days_Until_Year_End FROM Rentals;
```

Output:

rental_id	rental_date	days_until_year_end
1	2025-12-01	30
2	2025-12-10	21
3	2025-12-12	19
4	2025-12-16	15
5	2025-12-20	11
6	2025-12-21	10
7	2025-12-26	5
8	2025-12-27	4

4. Handle NULL return dates by displaying 'Not Returned Yet'.

```
1 SELECT rental_id, COALESCE(CAST(return_date AS VARCHAR), 'Not  
   Returned Yet') AS Return_Status FROM Rentals;
```

Output:

rental_id	return_status
1	2025-12-05
2	Not Returned Yet
3	2025-12-15
4	2025-12-18
5	Not Returned Yet
6	2025-12-25
7	Not Returned Yet
8	2025-12-29

5. Retrieve full rental contract details (Customer Name, Car Model, and Staff Name).

```
1 SELECT R.rental_id, C.full_name, Ca.model, S.staff_name  
2 FROM Rentals R  
3 JOIN Customers C ON R.customer_id = C.customer_id  
4 JOIN Cars Ca ON R.car_id = Ca.car_id  
5 JOIN Staff S ON R.staff_id = S.staff_id;
```

Output:

rental_id	full_name	model	staff_name
1	Sarah Nasser	Camry	Sara Ahmed
2	Sufyan Rashad	Tucson	Omar Khaled
3	Majd Nasr	S-Class	Fahad Ibrahim
4	Ramez Al-Omari	Camry	Omar Khaled
5	Omar Al-Kaff	Patrol	Sara Ahmed
6	Laila Al-Harbi	Accord	Omar Khaled
7	Sarah Nasser	Series 7	Fahad Ibrahim
8	Zaid Al-Sultan	488	Omar Khaled

6. List all cars along with their category name and daily rate.

```
1 SELECT Cars.model, CarCategories.category_name, CarCategories.  
   daily_rate  
2 FROM Cars  
3 JOIN CarCategories ON Cars.category_id = CarCategories.  
   category_id;
```

Output:

model	category_name	daily_rate
Camry	Sedan	50.00
Tucson	SUV	85.00
Series 7	Luxury	250.00
Accord	Sedan	50.00
S-Class	Luxury	250.00
Patrol	SUV	85.00
Rio	Hatchback	40.00
488	Sports	300.00
Transit	Van	100.00
Corolla	Sedan	50.00

7. Find cars that have never been rented using a Left Join.

```
1 SELECT Cars.brand, Cars.model  
2 FROM Cars  
3 LEFT JOIN Rentals ON Cars.car_id = Rentals.car_id  
4 WHERE Rentals.rental_id IS NULL;
```

Output:

brand	model
Kia	Rio
Ford	Transit
Toyota	Corolla

8. Calculate the total revenue collected for each payment method.

```
1 SELECT payment_method, SUM(amount) AS Total_Revenue
2 FROM Payments
3 GROUP BY payment_method;
```

Output:

payment_method	total_revenue
Apple Pay	600.00
Debit Card	100.00
Credit Card	450.00
Cash	750.00

9. Identify the staff member who processed the highest number of rentals.

```
1 SELECT staff_id, COUNT(*) AS Total_Rentals
2 FROM Rentals
3 GROUP BY staff_id
4 ORDER BY Total_Rentals DESC LIMIT 1;
```

Output:

staff_id	total_rentals
3	4

10. List car categories with an average daily rate greater than \$100.

```
1 SELECT category_name, AVG(daily_rate)
2 FROM CarCategories
3 GROUP BY category_name
4 HAVING AVG(daily_rate) > 100;
```

Output:

category_name	avg
Sports	300.000000000000000000
Luxury	250.000000000000000000

11. Count the number of currently available cars in each category.

```
1 SELECT category_id, COUNT(*)
2 FROM Cars
3 WHERE status = 'Available'
4 GROUP BY category_id;
```

Output:

category_id	count
1	2
2	2
4	1
5	1
6	1

12. List customers who made a payment higher than the overall average payment amount.

```
1 SELECT full_name FROM Customers
2 WHERE customer_id IN (
3     SELECT customer_id FROM Rentals
4     WHERE rental_id IN (SELECT rental_id FROM Payments WHERE
5         amount > (SELECT AVG(amount) FROM Payments))
6 );
```

Output:

full_name
Majd Nasr
Zaid Al-Sultan

13. Find the cars belonging to the category with the highest daily rate.

```
1 SELECT brand, model FROM Cars
2 WHERE category_id = (SELECT category_id FROM CarCategories WHERE
3     daily_rate = (SELECT MAX(daily_rate) FROM CarCategories));
```

Output:

brand	model
Ferrari	488

14. Generate a combined list of all individuals (Staff and Customers) in the system.

```
1 SELECT staff_name AS Name, 'Staff' AS Type FROM Staff
2 UNION
3 SELECT full_name AS Name, 'Customer' AS Type FROM Customers;
```

Output:

name	type
Sarah Nasser	Customer
Mona Ali	Staff
Hanan Mansour	Customer
Laila Al-Harbi	Customer
Zaid Al-Sultan	Customer
Sara Ahmed	Staff
Sufyan Rashad	Customer
Majd Nasr	Customer
Omar Al-Kaff	Customer
Noura Al-Said	Customer
Omar Khaled	Staff
Ali Hassan	Staff
Fahad Ibrahim	Staff
Layla Mohammed	Staff

15. List cars with model years between 2021 and 2024.

```
1 SELECT brand, model, year FROM Cars WHERE year BETWEEN 2021 AND 2024;  
2
```

Output:

brand	model	year
Toyota	Camry	2022
Hyundai	Tucson	2023
BMW	Series 7	2021
Mercedes	S-Class	2024
Nissan	Patrol	2022
Kia	Rio	2021
Ferrari	488	2023
Toyota	Corolla	2022

16. Search for customers whose names start with the letter 'S'.

```
1 SELECT * FROM Customers WHERE full_name LIKE 'S%';
```

Output:

customer_id	full_name	license_number	phone
101	Sarah Nasser	LIC-99901	0551234567
102	Sufyan Rashad	LIC-88802	0509876543

17. Display the most recent 5 payment transactions.

```
1 SELECT * FROM Payments ORDER BY payment_date DESC LIMIT 5;
```

Output:

payment_id	rental_id	amount	payment_date	payment_method
5	8	600.00	2025-12-29	Apple Pay
4	6	200.00	2025-12-25	Credit Card
3	4	100.00	2025-12-18	Debit Card
2	3	750.00	2025-12-15	Cash
1	1	250.00	2025-12-05	Credit Card

6 Conclusion:

This project demonstrates the design and implementation of a relational database for a car rental business. By utilizing PostgreSQL on Supabase, we established a system that ensures data integrity and supports complex querying to assist management in tracking fleet performance, customer activity, and financial revenue.